bipl-lsp

Editing BIPL, properly.

An LSP language server for a language from the Software Languages Book.

factorialV1.bipl — softlang/yas

```
1  while (i <= x) {   x : int · inferred program input
2    y = y * i;
3  }
```

The claim: even five statements hide two real language problems.

Vivekananda Reddy Godala SLE Assignment 3 · Option LSP.2 · Prof. Ralf Lämmel · 2026-07-13

The assignment — and what I claimed

OPTION LSP.2

Build an LSP editor for a recurring language in the Software Languages Book (YAS).

> The bar the assignment sets
>
> “We need to see an ***interesting*** LSP server and client.”
>
> — R. Lämmel, assignment post

So the whole game is: pick a language where the analysis is genuinely non-trivial.

MY CLAIM · OLAT, 23 Jun

I chose **`BIPL`** — small, but with real static semantics worth tooling:

- **type consistency** — arithmetic vs. boolean; conditions must be boolean
- **definite assignment** — a variable shouldn't be read before it is assigned

“Diagnostics from an actual analysis — not keyword matching.”

✓ **Confirmed** “Yes, BIPL is a good choice.” — RL, 6 Jul

Approach: reuse the lecture's LSP setup

Same stack the LSP lecture demoed for FSML: a `pygls` server + a VS Code client, talking `JSON-RPC` over stdio.

One request on the wire

```
{ "jsonrpc": "2.0", "id": 7,

"method": "textDocument/hover",

"params": { "position": {"line":0,"character":12} }
}
```

“what's at line 0, column 12?” → the server answers from its analysis.

The session lifecycle (from the lecture)

- **Start** — `initialize → initialized`
  capability negotiation
- **Sync** — `didOpen · didChange`
  server tracks the document
- **Ask** — `hover · definition · completion`
  client requests answers
- **Push** — `publishDiagnostics`
  server reports problems
- **Act** — `codeAction · formatting`
  server proposes edits

What's mine: I reproduced that architecture and swapped in a much richer analysis for BIPL.

The idea: why BIPL is interesting to tool

“A trivial imperative programming language … a small fragment of C.” — softlang/yas

division.bipl

```
// Euclidean division
{
  q = 0;
  r = x;
  while (r >= y) {
    r = r - y;
    q = q + 1;
  }
}
```

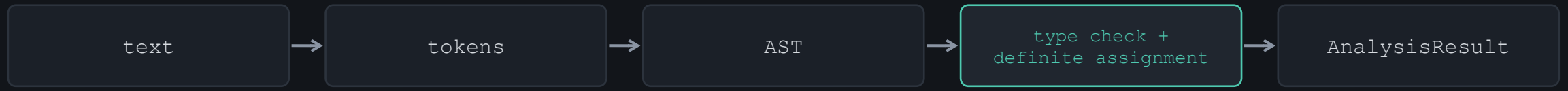
Three facts make it interesting to tool

No declarations. You never write int x. The editor must infer every type — nobody states it.

No boolean literals. No true / false keyword. Booleans exist only as results of comparisons and && || !.

Inputs live in the store. No input statement — a program just reads variables already in memory. That is why samples read x without assigning it.

Under the hood: one analysis, real static semantics



Type checking

types.bipl

```
1  x = 5;
2  x = 1 < 2;   int can't become bool
```

CHAPTER 9 · WELL-TYPEDNESS

`m ⊢ e : T` in context m, expression e has type T

$$\frac{e0 : bool \qquad e1 : T \qquad e2 : T}{\texttt{if (e0) e1 else e2} \; : \; T}\;[t\text{-}if]$$

Our checker implements this — and completes the operators the YAS Haskell left as "TODO".

Definite assignment

A read is safe only if the variable is assigned on every path. An if needs both branches; a while body may run zero times. (A data-flow pass — beyond Ch. 9.)

flow.bipl

```
1  if (c) y = 1;
2  z = y;   y maybe uninitialized
3
4  while (c) v = 1;
5  u = v;   v maybe uninitialized
```

One result, every feature. Hover, definition, rename, completion and the outline are all just queries on this.

Not every unassigned read is a bug

BIPL reads its inputs from the store. So a naive “read-before-write = error” rule would flag `factorialV1.bipl` — an official sample — as broken. **The language would reject its own textbook.**

PROBLEMS

✖ Error	`type-error`	conditions must be boolean; a variable can't change type
⚠ Warning	`maybe-uninitialized`	assigned somewhere, but not on every path to this read
ⓘ Information	`input-variable`	never assigned anywhere → a program input; this is fine

12/12

official YAS samples
analyse with zero errors

— and a test locks it in,
so it can never regress.

Respecting the language's own semantics — instead of imposing a naive rule — is what makes the server “interesting.”

Demo — in action

bad.bipl

```
1  {
2    x = 5;
3    if (x) y = 1;   condition must be boolean
4    z = y + 1;   y maybe uninitialized
5    w = (2 < 3) + 4;   '+' expects integers
6  }
```

PROBLEMS ✖ 3 errors ⚠ 2 warnings

1. **good.bipl**
   blue input hints; hover shows types inferred from use
2. **bad.bipl**
   3 type errors + 2 data-flow warnings, live
3. **Navigate**
   hover · go-to-definition · find references
4. **Refactor**
   F2 rename (rejects invalid names) · quick fix
5. **Live typing**
   break factorial, watch diagnostics move as you type

Challenges & loose ends

What I had to solve — and what I deliberately left open.

✓ Solved

Inputs vs. bugs. A naive rule broke the samples → three severities + an “assigned-anywhere” pre-pass.

Types without declarations. Demand-driven inference: a constrained use (i <= x) pins the type.

Completed the checker. The YAS TypeChecker.hs ended at “TODO”; I implemented the full operator set.

Stayed useful while broken. Error recovery: skip to the next ; or } so half-typed code still analyses.

○ Left open (on purpose)

No full type unification. A variable used only neutrally (r = x) stays unknown — I infer from constrained uses only.

Re-analysis per keystroke. Whole-document, not incremental. Linear and instant at BIPL's scale; wouldn't scale to big files.

Fidelity vs. usability. Relaxed the Haskell checker's both-branches-equal rule, and accept the conventional identifier shape (a superset of ls.term).

Three things to remember

1 **LSP decouples smartness from the editor.**
One Python analysis serves any LSP editor; the VS Code client is 30 lines.

2 **A small language is not trivial tooling.**
Five statements still forced type inference without declarations and branch-sensitive data flow.

3 **The language's own samples were the spec.**
They told me inputs aren't errors — and now they guard that decision as tests. This is the FSML lecture demo, extended to a harder language.

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code · tests · demo files `github.com/VivekanandaReddy3/sle-bipl-lsp`